

Rapid Deployment of Antiviral Drugs Using Single-Virus Tracking and Machine Learning

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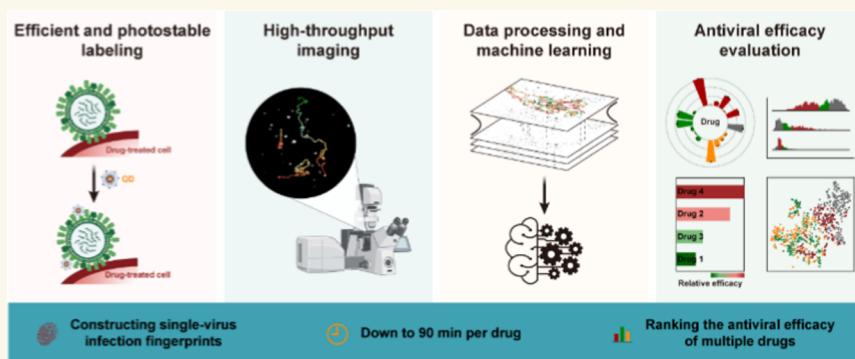
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ABSTRACT: The outbreak of emerging acute viral diseases urgently requires the acceleration of specialized antiviral drug development, thus widely adopting phenotypic screening as a strategy for drug repurposing in antiviral research. However, traditional phenotypic screening methods typically require several days of experimental cycles and lack visual confirmation of a drug's ability to inhibit viral infection. Here, we report a robust method that utilizes quantum-dot-based single-virus tracking and machine learning to generate unique single-virus infection fingerprint data from viral trajectories and detect the dynamic changes in viral movement following drug administration. Our findings demonstrated that this approach can successfully identify viral infection patterns at various infection phases and predict antiviral drug efficacy through machine learning within 90 min. This method provides valuable support for assessing the efficacy of antiviral drugs and offers promising applications for responding to future outbreaks of emerging viruses.

KEYWORDS: antiviral drugs, drug repurposing, machine learning, single-virus tracking, quantum dots

INTRODUCTION

The emergence and spread of viruses pose a significant threat to human health and the global socio-economic landscape, resulting in a substantial economic and social security burden.¹ Accelerating the development of antiviral drugs can provide faster and more effective treatments, reducing the spread and impact of the virus. Despite rapid advancements in modern scientific research, drug development often faces challenges, such as long development cycles and high costs. Reusing approved drugs or candidate drugs that have already undergone clinical evaluation is a feasible strategy to shorten the drug development cycle.² This approach offers many advantages compared to developing drugs for specific indications, such as reducing the risk of failure, shortening drug development time, and decreasing investment.³ Currently, this strategy has been used for antiviral drug screening against Zika,⁴ Ebola,⁵ hepatitis C,⁶ influenza,⁷ dengue viruses,⁸ and SARS-CoV-2.^{9,10} There-

fore, it is crucial to use drug repurposing methods to identify antiviral drugs with repurposing potential.

Currently, there are two primary approaches to drug repurposing. The first method is knowledge-based repurposing, which relies on an understanding of disease-related pathways,^{11,12} drug molecular structures,¹³ or target mechanisms¹⁴ for drug screening. The second approach does not require knowledge of targets and disease mechanisms but instead screens phenotypes directly. For acute infectious diseases like COVID-19, the rapid use of knowledge-based methods is

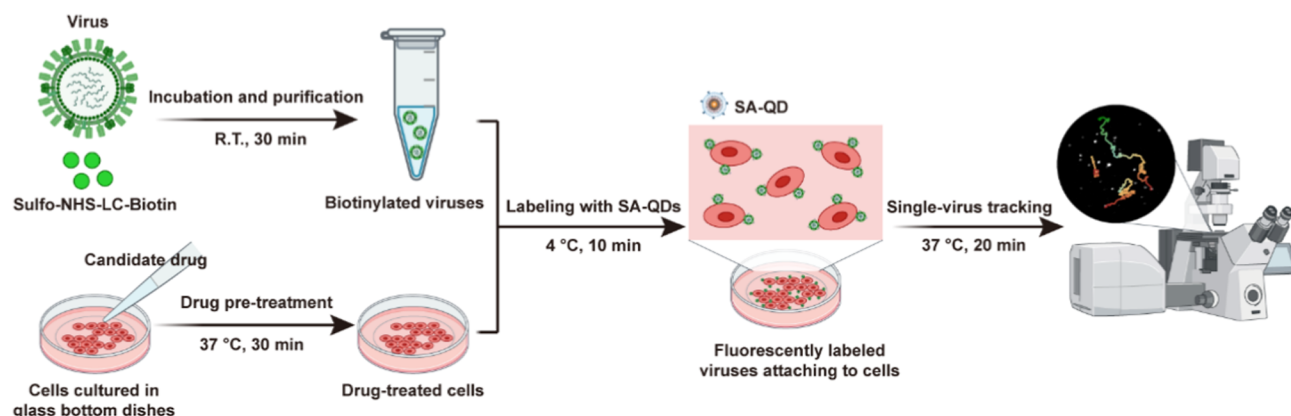
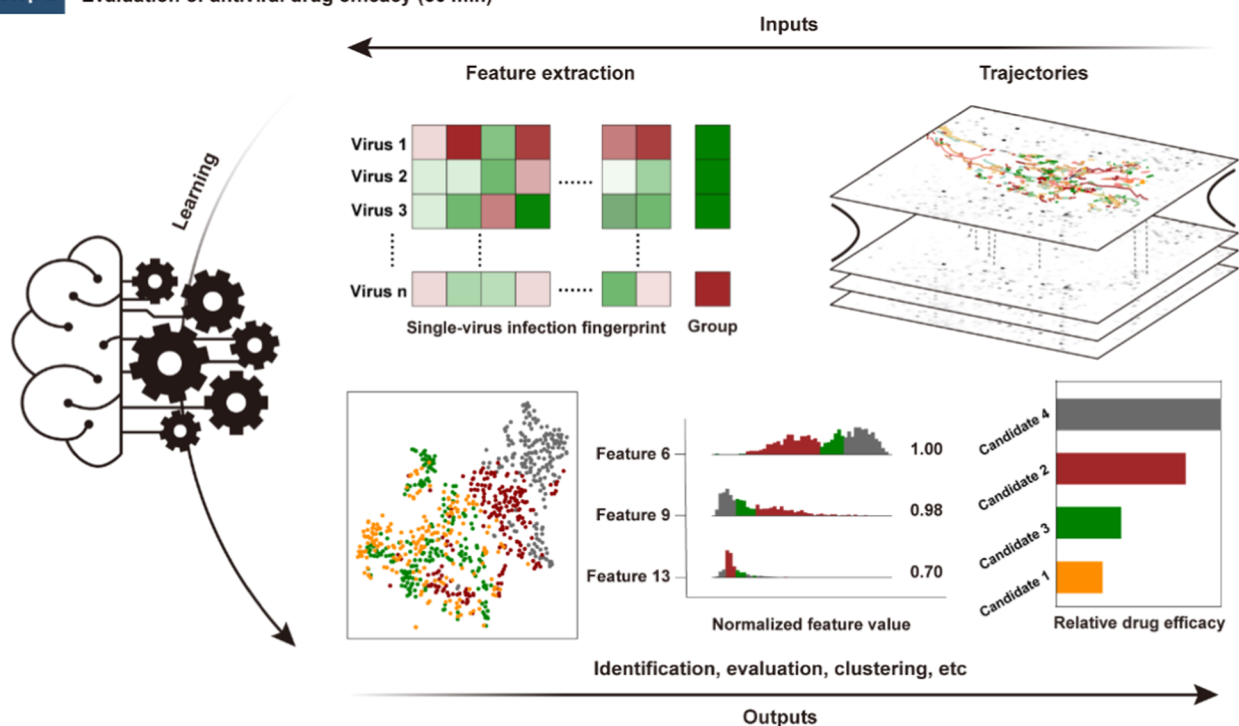
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Scheme 1. Machine Learning Based on SVIF for Rapid Evaluating and Ranking the Antiviral Drug Efficacy^a**Step 1** High-throughput and high-fidelity single-virus tracking (60 min)**Step 2** Evaluation of antiviral drug efficacy (30 min)

^aStep 1: high-throughput and high-fidelity SVT via an efficient, rapid, and photostable labeling technique. SA-QDs were employed to swiftly label biotinylated viruses attached to cell membranes during cell sample preparation. Subsequently, time series images capturing viral movements in live cells were acquired to extract infection trajectories under specific conditions. Step 2: evaluation of antiviral drug efficacy using machine learning based on the SVIF. As input sets, trajectories were reconstructed and analyzed to extract 17 descriptive features to obtain the SVIFs under corresponding treatment. By using the logistic regression model, the efficacy of antiviral drugs can be identified and evaluated based on fingerprint difference between the drug-treated and the control group.

limited due to a lack of understanding of emerging viruses. Therefore, phenotypic screening often helps to speed up the antiviral drug screening process. At present, phenotypic screening mainly utilizes host cells of the target virus to interact with the candidate compounds and then assesses the antiviral effects of the candidate molecules by measuring cell viability,^{15,16} pathway-related key molecular content,⁴ or viral replication levels^{9,10} after viral infection. While these traditional phenotypic screening methods have been instrumental in screening drugs against emerging viruses, they often require cumbersome steps for simultaneous validation of various aspects

of viruses and cells and lack direct visual evidence that drugs can prevent viral infection. In addition, for emerging viruses with high lethality and high transmission rates, the traditional phenotypic screening cycle still needs to be shortened. Therefore, there remains an urgent need to develop fast and visual phenotypic screening methods for antiviral drugs to rapidly identify candidate drugs.

Viral infection of host cells is a highly complex process involving different infection pathways and complex interactions between viruses and cellular structures.^{17–19} Our group has been focusing on a single-virus tracking (SVT) technique based on

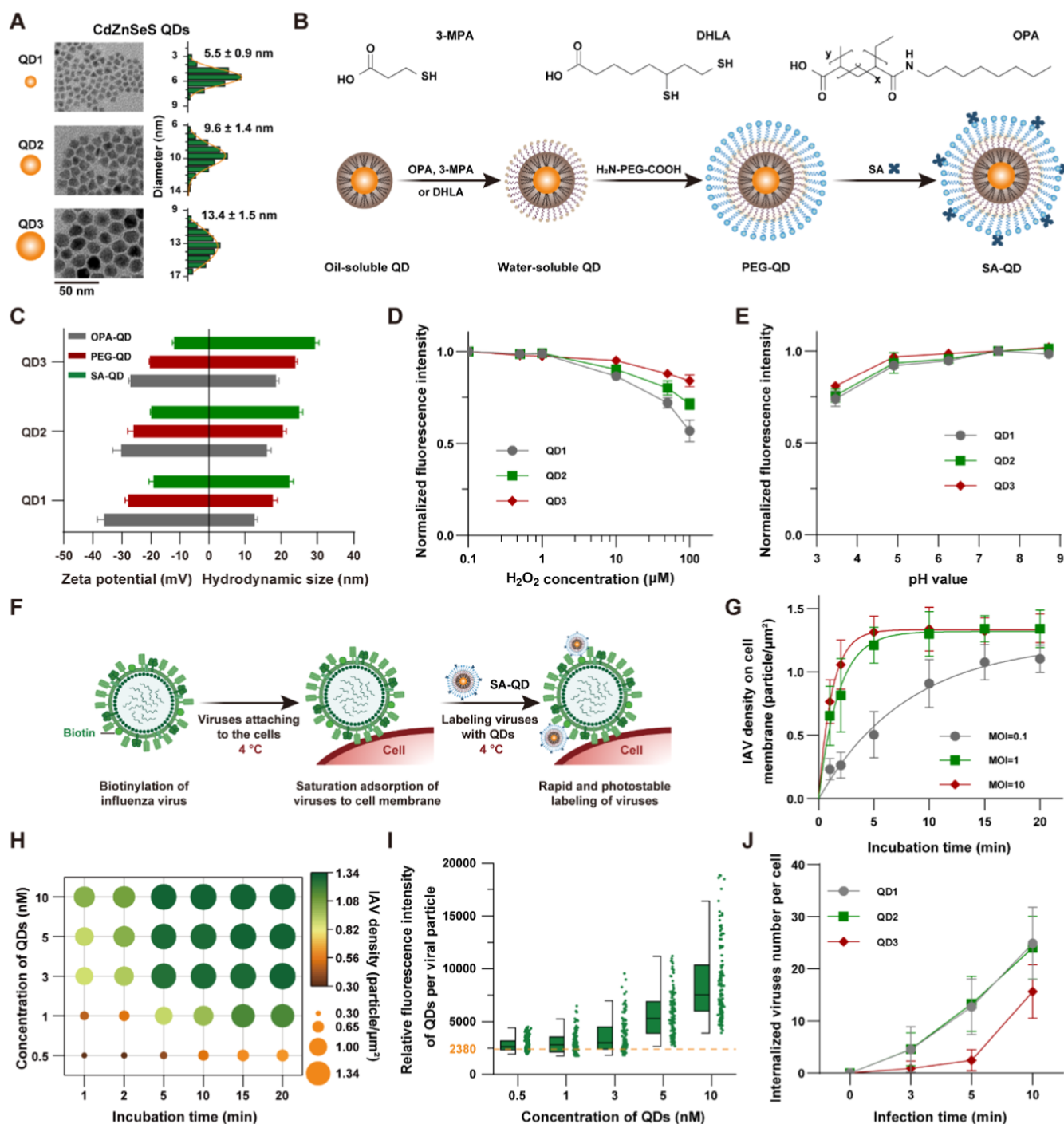


Figure 1. Efficient, stable, and interference-free labeling of viruses using QDs. (A) Transmission electron microscope images and corresponding particle diameter distributions of three oil-soluble CdZnSe QDs. (B) Schematic illustration of the procedure to prepare SA-QDs. Initially, the CdZnSe QDs were rendered water-soluble through modification with 3-mercaptopropionic acid (3-MPA), dihydrolipoic acid (DHLA), or octylamine-grafted poly(acrylic acid) (OPA), followed by conjugation with poly(ethylene glycol) (PEG) to enhance biocompatibility, and ultimately modified with streptavidin (SA). (C) Hydrodynamic size and zeta potential of differently modified QDs shown in (B). (D,E) Examination of SA-QDs for reactive oxygen species and acid resistance. Relative fluorescence intensity of SA-QDs were measured after dispersion in H₂O₂ solution or acidic solution for 2 h. (F) Schematic diagram of the specific steps to obtain QDs-labeled viruses binding to the surface of host cells. (G) Binding curve of Syto 82-labeled virus attaching to cell membrane. (H) Bubble heat map of the correlation between the incubation time, the concentration of SA-QDs, and the virus labeling efficiency. Color bar and bubble size indicate the QDs-labeled virus density on membrane from 0.30 (sparse) to 1.34 (saturated). (I) Relative fluorescence intensity of individual biotinylated virus particles after incubation with the indicated concentrations of SA-QDs. Orange dashed line represents the average fluorescence intensity of individual SA-QDs under the same conditions. (J) Amount of internalized QDs-labeled virus at different time post-infection. Box chart center line indicates median, the upper and lower box range 25th and 75th percentile, and the whiskers 1.5 × IQR. Bar chart and line chart, mean ± SD.

quantum dots (QDs) for a long time, which allows real-time tracking of the infection behavior of hundreds or thousands of

single viruses in live cells at single-virus resolution.^{20–22} SVT enables high-throughput quantitative analysis of the infection

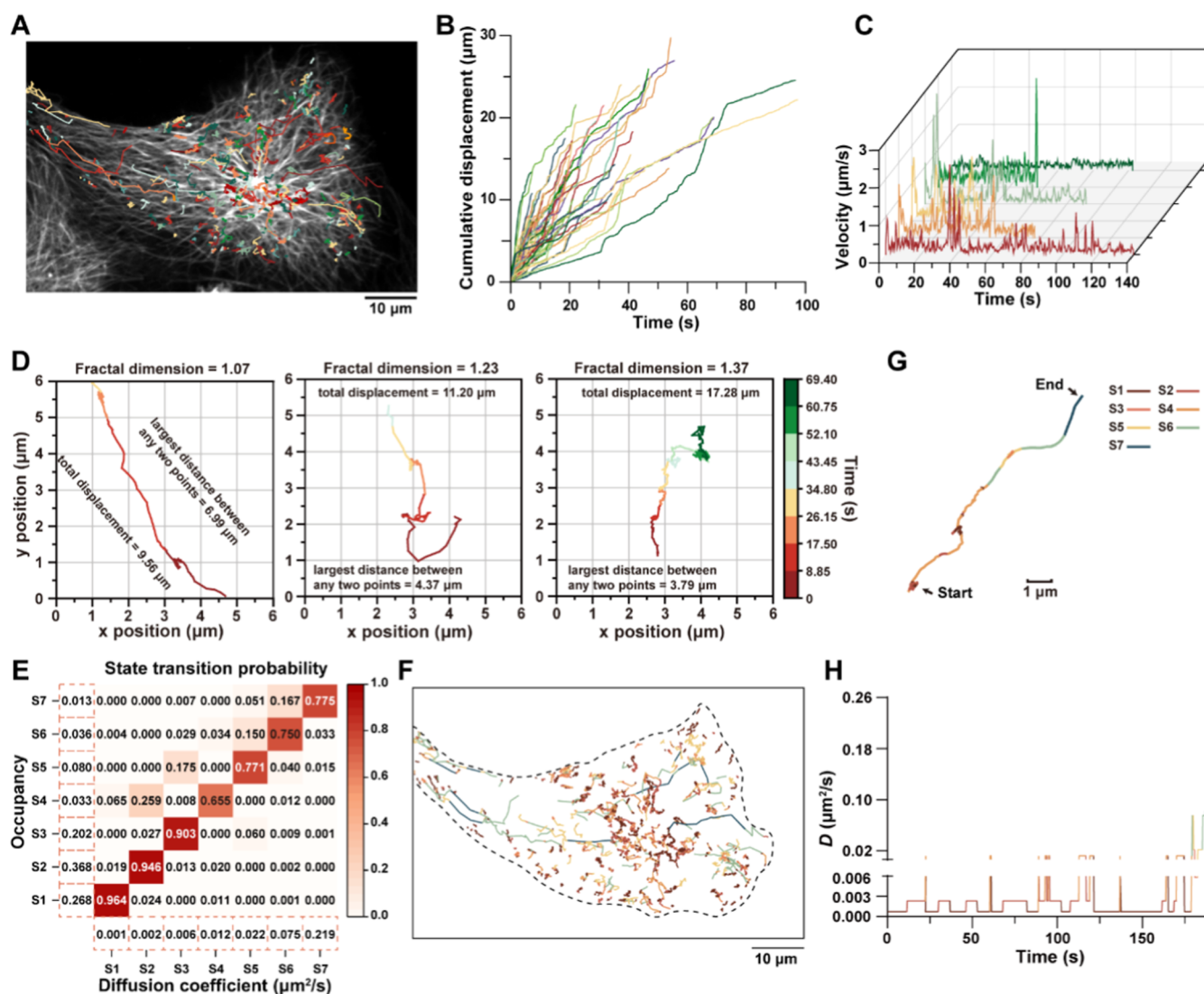


Figure 2. Heterogeneity of viral transport in live MDCK cells. (A) Infection trajectories of single viruses in live cells within 20 min of infection. (B) Cumulative displacement over time for 57 viral transport trajectories collected from 5 MDCK cells. (C) Instantaneous velocity over time for 6 trajectories randomly selected from (B). (D) Three representative trajectories with varying degrees of direction deviation showing similar fractal dimension values with color indicating time. (E) State occupancy, state transition probability, and diffusion coefficients derived from the hidden Markov model (HMM) analysis. (F–H) HMM analysis of viral transport states. HMM analysis depicts viral transport trajectories in (A), and a representative trajectory (G) with the result of the global HMM state changes (H), illustrating seven states (S1–S7, in different colors). 2643 viral trajectories from healthy MDCK cells were collected for HMM analysis.

trajectories of individual viruses with nanometer spatial resolution and millisecond temporal resolution, thereby dissecting the interactions between viruses and host cells that are masked by ensemble averaging.^{23–25} Since environmental disturbances and the structural and functional interactions between viruses and host factors can affect the viral infection process and result in changes in viral transport characteristic parameters,^{26–28} the dynamic infection characteristics extracted by SVT could serve as valuable “fingerprint” information.²⁹ Additionally, machine learning methods have significant advantages in handling, integrating, and analyzing massive amounts of data information. In recent years, the combination of single particle or molecule tracking with machine learning has been successfully used to explore biological data trends and make predictions.^{30–33} Therefore, the integration of SVT and machine learning holds promise as a rapid, visual antiviral drug

phenotypic screening method to achieve rapid and urgent evaluation of the antiviral efficacy of clinically candidate drugs.

In this study, we developed a fast and visual drug screening technique using SVT and machine learning with the influenza virus as a model. Thanks to the high brightness and excellent photostability of QDs, this approach enables quick and accurate prediction of antiviral drug effectiveness within 90 min through the integration of two main components: high-fidelity and high-throughput QD-based SVT and high-accuracy evaluation of antiviral drug efficacy with machine learning. First, we treated the host cells with drugs and the virus with biotinylation reagent for 30 min, respectively. Then, the biotinylated viruses were attached to the host cell membrane and labeled with streptavidin-modified QDs (SA-QDs) to obtain the cell samples bound to QDs-labeled viruses for imaging. By continuously capturing images of virus-infected cells over 20 min and performing precise trajectory extraction and analysis, we

gathered detailed information about thousands of single-virus infection trajectories. By extracting unique single-virus infection fingerprint (SVIF) data from viral trajectories, we successfully identified viral infection patterns at various phases and predicted the efficacy of antiviral drugs through machine learning (Scheme 1). This method is poised to serve as a robust tool for swift screening of antiviral medications for emerging viruses in the future.

RESULTS AND DISCUSSION

Efficient, Rapid, and Photostable Labeling of Viruses Attached to Host Cell Membrane. QDs are excellent tools for tracking viruses in living cells due to their exceptional fluorescent properties and photostability. To label viruses effectively without affecting their biological activity, the QDs need to exhibit good colloidal stability, high brightness, and biocompatibility and be small enough to avoid interfering with viral infection. In our study, we synthesized three sizes of CdZnSeS QDs with diameters of 5.5 ± 0.9 nm, 9.6 ± 1.4 nm, and 13.4 ± 1.5 nm, which displayed well-defined shapes, uniform size distribution, and a high fluorescence quantum yield (Figures 1A and S1). We modified them using 3-mercaptopropionic acid (3-MPA), dihydrolipoic acid (DHLA), or octylamine-modified poly(acrylic acid) (OPA), thereby improving their water solubility. Among them, OPA-modified QDs (OPA-QDs) showed better water stability (Figures 1B and S2). The introduction of poly(ethylene glycol) (PEG) onto the OPA-QDs (PEG-QDs) improved biocompatibility and reduced the nonspecific adsorption of QDs. SA was covalently linked to PEG-QDs (SA-QDs) for specific labeling of biotinylated viruses (Figure 1C). Challenges such as elevated reactive oxygen species levels and decreased endosomal pH existed during viral infection, and fluorescence intensity analysis indicated that the larger-sized SA-QDs were able to maintain stable fluorescence intensities under these conditions, whereas the smaller-sized SA-QDs had relatively weaker antioxidant capacity (Figure 1D,E).

To track individual viruses within living cells over a long period of time with efficiency and accuracy, we used a two-step method involving the biotin–SA interaction to label viruses with QDs (Figure 1F).³⁴ After treating the viruses with biotinylated reagent for 30 min to align with the time when most antiviral drugs take effect in cell culture, we infected the MDCK cells with biotinylated viruses at a multiplicity of infection (MOI) of 1 for 5 min, enabling high-throughput monitoring of viral mobility (Figure 1G). Subsequently, we labeled the viruses with SA-QDs at a concentration of 3 nM or higher in 5 min, achieving an efficiency exceeding 95% (Figure 1H). Further analysis of fluorescence intensity on individual viruses and transmission electron microscopy (TEM) images revealed a positive correlation between the number of QDs bound to individual viruses and the concentration of SA-QDs, with most virus particles displaying 1–3 QDs bound at a concentration of 3 nM for SA-QDs (Figures 1I and S3). Due to the high fluorescence intensity and stability, a single QD is sufficient for long-term tracking. To minimize interference with viral infectivity, we selected 3 nM SA-QDs and a 5 min incubation time for labeling influenza viruses. Further experiments observing that the 13.6 nm QD-labeled viruses experienced a delayed entry into the cells (Figures 1J and S4). Collectively, the 9.6 nm QDs did not interfere with virus infection and exhibited superior photostability, making them the preferred choice for subsequent experiments (Figures S5 and S6A).

Under these optimized experimental conditions, we also labeled the viral genome with the nucleic acid dye Syto 82 to verify the effectiveness of the QD-labeling method. Colocalization analysis showed that nearly all QD signals colocalized with Syto 82 signals, with a colocalization efficiency exceeding 90% (Figure S6B–E). The photostability assessment confirmed that QD signals remained stable throughout the tracking period without any noticeable blinking or bleaching (Figure S6F). These results indicated that our labeling strategy can meet the requirements for long-term high-fidelity tracking of individual viruses. Imaging results at different frame rates showed that a frame rate of 4.10 Hz provided detailed viral movement without compromising trajectory resolution compared to 8.20 Hz while maintaining a high signal-to-noise ratio (Figure S7). This frame rate was deemed to be suitable for high-fidelity imaging without significantly increasing data processing complexity.

Tracking Individual Viruses in Live Cells with High Throughput. We then placed the prepared samples on a confocal microscope equipped with a live-cell culture system to monitor the infection process of the individual viruses in live cells. Viruses initially stayed on the cell surface at 0 min post-infection, then moved into the cytoplasm, forming aggregates near the nucleus. By recording the viral transport trajectories within 20 min post-infection, we observed that many influenza viruses move nearly linearly from the cell membrane to the perinuclear region using host transport systems (e.g., microtubules), but different virus subpopulations exhibit varying degrees of distance and directional deviation (Figure 2A). Subsequently, we investigated the cumulative displacement and instantaneous velocity changes of multiple viruses over time. Cumulative displacement curves were stepped, and instantaneous velocity curves were multi-peaked, suggesting the existence of alternating fast and slow states during viral infection (Figure 2B,C). To measure the differences in the direction of movement between virus subpopulations, we used the fractal dimension to quantify the degree of linearity of the viral movement trajectories. The analysis showed that despite the presence of directional shifts of more than 90° in the viral motions, the movements still conformed to the linearity from an overall perspective (Figure 2D).

To analyze the fast and slow movement states of viruses in more detail, we analyzed all the viral movement trajectories during the infection period by using the HMM.³⁵ The HMM classified the viral movement into seven states based on their instantaneous speeds and calculated the proportion of each state in the whole movement process as well as the transition probabilities between the states. The results showed that states with diffusion coefficient less than $0.01 \mu\text{m}^2/\text{s}$ accounted for more than 80% of the viral motions (Figure 2E). The HMM state transitions of typical trajectories indicated a correlation between state changes and the local cellular environment, where virus particles were located (Figure 2F–H). By tracking the viruses within the cells in which microtubules were fluorescently labeled, we found that the complexity of microtubules could lead to fluctuations in viral velocity (Figure S8). In addition, previous studies have shown that vesicles transporting viruses undergo velocity changes during viral maturation.^{26,34} Thus, changes in the local cellular environment and alterations in the structure and function of the virus can affect the HMM state. Overall, the transport process of viruses is complex and diverse, and despite the heterogeneity of transport properties among individual virus particles, the transport properties of virus populations are predictable under the same experimental conditions.

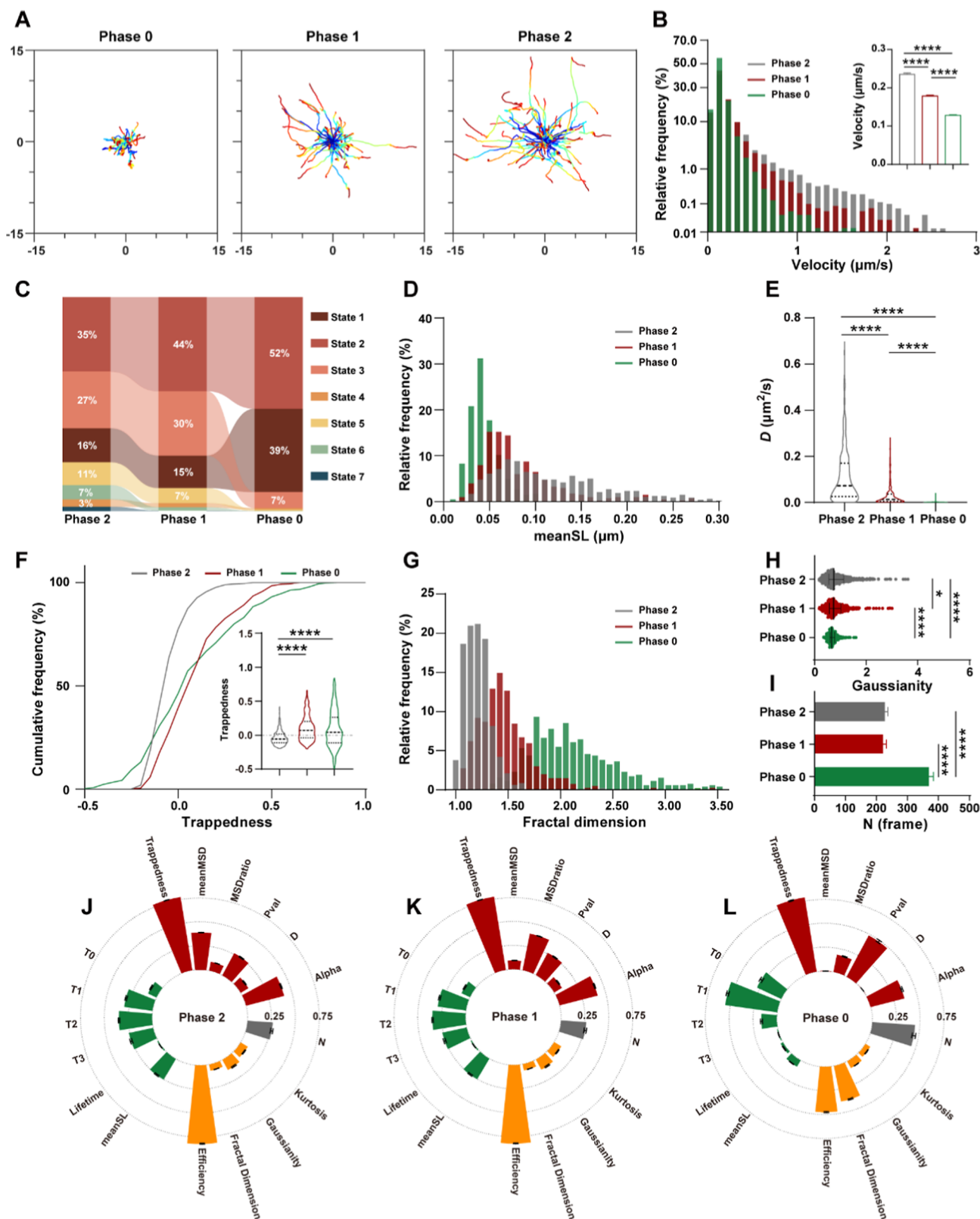


Figure 3. Mapping SVIFs of the three infection phases. (A) Representative trajectories in phase 0 (infection for 0–2 min, $n = 61$), phase 1 (infection for 2–5 min, $n = 67$), and phase 2 (infection for 5–20 min, $n = 58$). Unit, μm . (B) Instantaneous velocity means and distribution profiles of the three infection phases. (C) Variation of HMM states occupancy varied with infection phase. (D,G) Distribution profiles of mean step length (meanSL) and fractal dimension of the three infection phases. (E,F,H,I) Comparison of the diffusion coefficient (D), trappedness, Gaussianity, and persistent frame number (N) in the three phases of viral infection. (J–L) Radial bar graph showing the SVIFs in the three phases of viral infection. Each phase used approximately 400 trajectories for calculating features. Violin plot center line indicates median, and the upper and lower line 25th and 75th percentile. Bars, mean $\pm$ SEM. Statistical analysis was performed by one-way ANOVA. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$.

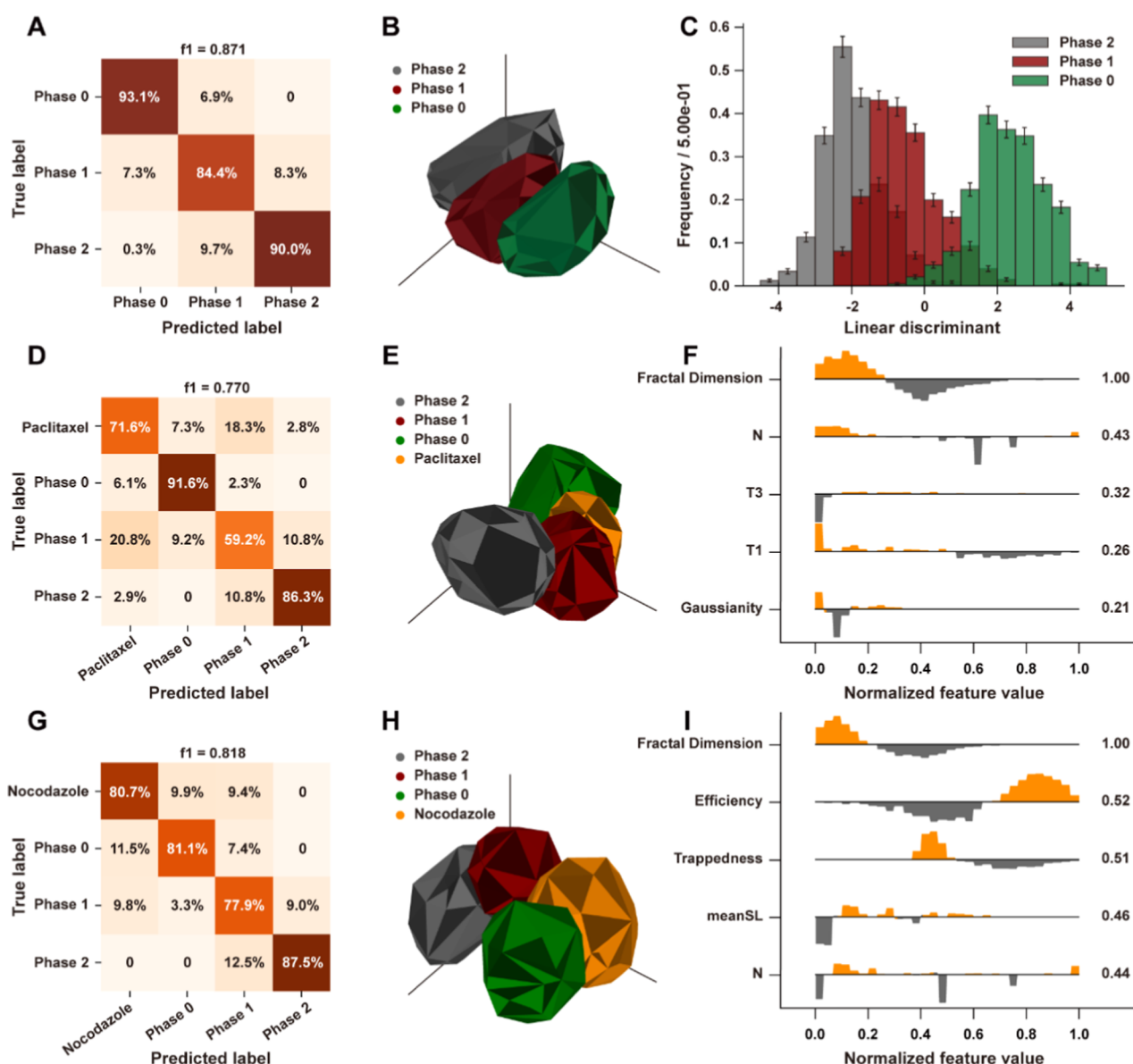


Figure 4. Classification and feature ranking for paclitaxel- and nocodazole-treated groups via machine learning. (A,D,G) Confusion matrix for prediction using a logistic regression model trained to separate the fingerprints. (B,E,H) Three-dimensional PCA projection of the SVIFs. Spheres represent the convex hull containing 1σ of the data from the mean for each group. (C) One-dimensional LDA projection of the SVIFs for the three phases. (F,I) Normalized distribution of the five most descriptive features in the SVIFs based on a ranking of components in a one-dimensional LDA projection. Histogram bins are colored based on the variant with highest value for that bin. Numbers denote the normalized vector component in the LDA projection.

Mapping Single-Virus Infection Fingerprints (SVIFs) with 17 Features. As influenza virus infection progresses, we investigated variations in viral transport patterns across different infection phases. The viral infection process was segmented into three phases based on infection time, with 0–2 min denoted as phase 0, 2–5 min as phase 1, and 5–20 min as phase 2. Comparative analysis of viral trajectories and instantaneous velocities at distinct infection phases revealed a gradual increase in viral movement velocity as the infection time advanced (Figure 3A,B). The HMM state analysis further indicated a rise in the proportion of fast HMM states with prolonged infection time, accompanied by a corresponding increase in the median diffusion coefficient (phase 2 = $0.07 \mu\text{m}^2/\text{s}$, phase 1 = $0.01 \mu\text{m}^2/\text{s}$, and phase 0 < $0.001 \mu\text{m}^2/\text{s}$). This suggested that a subpopulation of viruses with large displacement and fast

movement dominated as infection time increased (Figure 3C–E). Moreover, despite significant differences in the range of viral motion observed across the three infection phases, similar median values of trappedness (phase 2 = -0.06 , phase 1 = 0.07 , and phase 0 = 0.04) indicated that viral motion remained oriented even when viruses were confined near the cell membrane during the early infection phases (Figure 3F).

The fractal dimension analysis illustrated an increasing trend in the linearity of viral motion with a prolonged infection time. Meanwhile, the feature Gaussianity, which also characterizes directionality, exhibited similar median values across the three infection phases (phase 2 = 0.74 , phase 1 = 0.67 , and phase 0 = 0.75). This suggested that there are differences in features characterizing viral movement from the same perspective, and that variability in viral movement needs to be characterized by

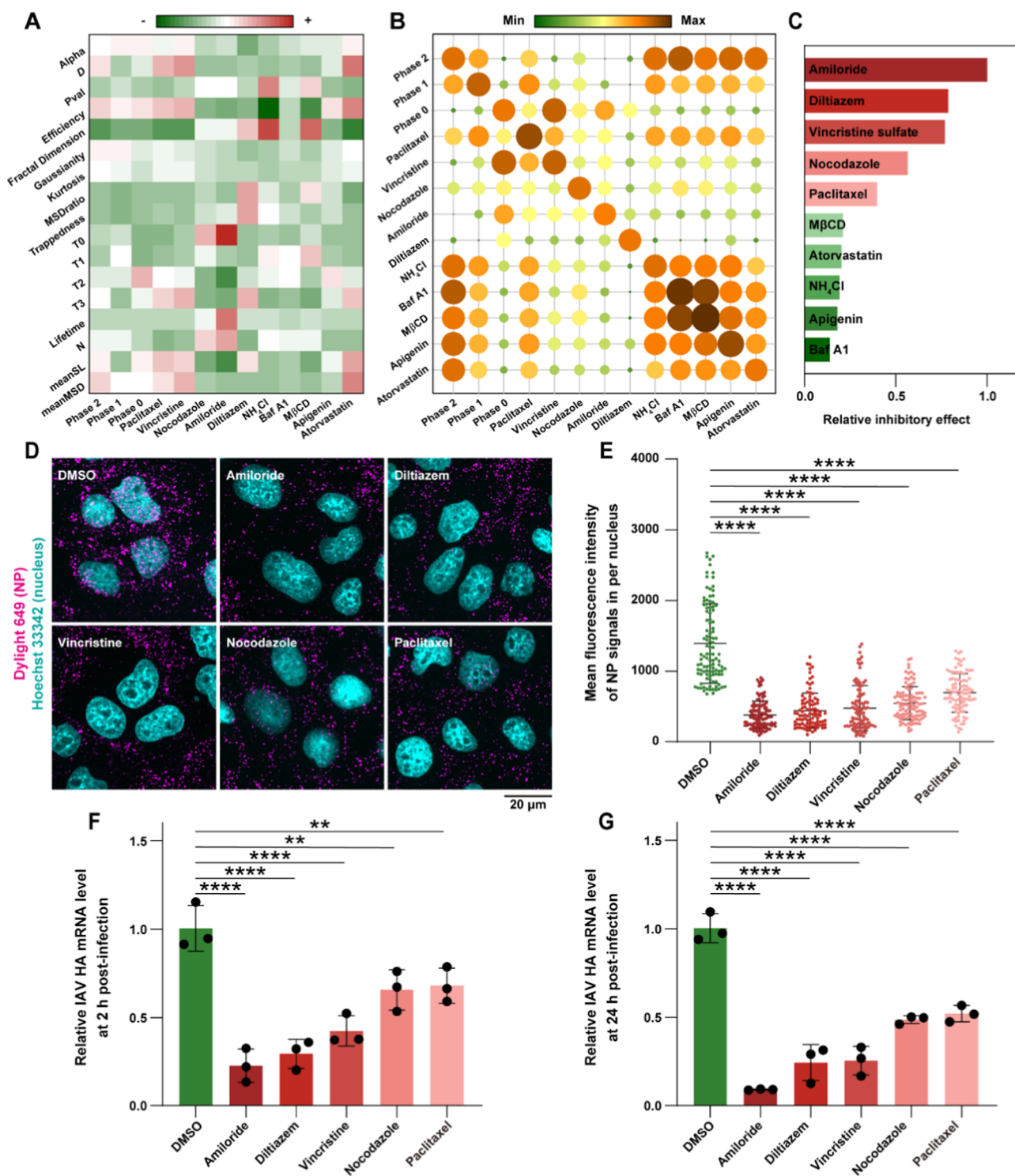


Figure 5. Evaluating and ranking the effects of different drugs on IAV infection by machine learning based on the SVIFs. (A) Heat map showing averaged feature patterns of viral infection trajectories from all groups. (B) Spatial proximity of the patterns under various drug treatment. Intra- and inter-group similarities were determined, respectively, by the average Euclidean distance of the SVIFs in each group or paired groups. In the similarity matrix, larger circles and green-to-red tendency represent higher similarity. (C) Ranking of interfering viral transport in groups treated with a specific concentration of compounds. Effectiveness of 10 molecules was judged by the phase 2 inter-group Euclidean distance of 0.21 as a baseline. (D) Representative images of MDCK cells immunofluorescence labeled for IAV nucleoprotein (NP; magenta) and nuclei (with Hoechst 33342; cyan) at 2 h post-infection. (E) Total fluorescence intensity of viral nucleoprotein in cell nucleus from compound-treated MDCK cells by immunofluorescence at 2 h post-infection. Center line indicates median, and the upper and lower line 25th and 75th percentile. (F,G) Virus production from compound-treated MDCK cells by real-time qPCR ($n = 3$ cultures) at 2 h (F) and 24 h (G) post-infection. Each tested group used 1800–2000 trajectories for machine learning. Values represent mean $\pm$ SD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$; statistical analysis was performed by a one-way ANOVA.

multiple methods for a comprehensive understanding (Figure 3G,H). Additionally, considering that the duration of viral motion N can reflect the likelihood of viral movement away from the 2D plane, the notably higher N value at phase 0 indicated that in the early phases of viral infection, the majority of viruses were constrained to the 2D plane near the cell membrane with limited movements. There was an increasing tendency for viruses to move away from the 2D plane in the Z -direction with an extended infection time (Figure 3I). Taken together, the transport properties of viruses varied with the degree of infection.

In order to highlight these differences more intuitively, we selected a total of 17 different features focusing on fluctuation in speed, spatial filling properties and linearity of transport trajectories, and duration of viral motion to establish SVIFs (Figure 3J–L, see Methods). By comparing the infection fingerprints of the three different infection phases, we can directly observe the differences in viral transport behavior due to changes in the biological environment caused by different infection levels.

Identifying Small Differences of the SVIFs by a Logistic Regression Model. To generate SVIFs under specific conditions, a dataset comprising at least 10^3 virus trajectories is required. The time-consuming and laborious processing, calculation, and analysis processes prompted the exploration of machine learning techniques. Leveraging machine learning enables swift creation, comparison, and analysis of SVIFs across various treatment groups by inputting XY coordinate information on trajectories and outputting the five features with the highest differentiation weight. In this study, a logistic regression classifier was trained by using SVIFs comprising 17 features. Subsequently, this classifier was employed to predict the three infection phases of influenza viruses with accuracies of 93.1, 84.4, and 90.0%, along with an F1 score of 87.1% (Figure 4A). Projections based on principal component analysis (PCA) and linear discriminant analysis (LDA) dimensionality reduction methods exhibited distinct inter-group distributions, showing the capability of the model to effectively analyze two groups of viral trajectories with highly overlapping step distributions, even for the phase 2 and phase 1 groups, and underscoring the robust performance of the model in analyzing complex and similar groups (Figure 4B,C).

Furthermore, machine learning was employed to identify subtle perturbations in viral behavior following drug treatment, serving as a basis for characterizing the antiviral efficacy of the drug. Two microtubule inhibitors, paclitaxel and nocodazole, were utilized to impede the viral infection process. The results from the confusion matrix and PCA 3D projection illustrated that treatment with paclitaxel and nocodazole impeded the large-scale and rapid infection movement of the virus, resulting in characteristics similar to those observed in the early infection phases (Figure 4D,E,G,H). By ranking the 17 descriptive features of the phase 2 group and the treatment groups based on discriminant weight, it was revealed that the most influential feature in both treatment groups was fractal dimension, indicating a loss of directionality in viral motion (Figure 4F,I). Further statistical analysis unveiled similarities and differences in transport features between the two inhibitor treatment groups (Figure S9). Potential interference from dimethyl sulfoxide (DMSO, 0.1 vol %) on viral transport was ruled out by quantifying all viral transport features (Figure S10). The outcomes of machine learning underscored its efficacy in

characterizing drug antiviral efficacy based on differences in SVIFs.

Evaluating the Efficacy of Antiviral Drugs by Machine Learning. Subsequently, we attempted to establish an antiviral drug screening platform utilizing SVIFs and machine learning, creating a comprehensive assessment system for evaluating the antiviral efficacy of various drugs. In this study, we tested 10 drugs, including microtubule inhibitors such as paclitaxel, vincristine sulfate, and nocodazole; endosomal acidification inhibitors NH_4Cl and Bafilomycin A1; the cholesterol-lowering agent methyl- β -cyclodextrin ($\text{M}\beta\text{CD}$); and four approved drugs with antiviral potential, diltiazem,³⁶ amiloride,³⁷ apigenin,³⁸ and atorvastatin,³⁹ which were originally used for other diseases. Cells were treated with optimized drug concentrations, and viral trajectories during 5–20 min of infection were extracted to construct SVIFs under certain conditions. The viral transport features under various molecular treatments were visually demonstrated through parallel plots (Figure S11).

Subsequently, characteristic fingerprints were generated by z -score normalization of the 17-feature matrix (Figure 5A). The control group fingerprints (phase 2, phase 1, and phase 0) exhibited a light color distribution, serving as a baseline for comparison with other groups. The strong green and red lumps in the drug-treated groups indicated the pharmaceutical effects on viral infection from various perspectives. By calculating the average Euclidean distance of the SVIFs in different group pairs, the antiviral effects of drugs could be visually discerned, and the mechanistic similarities between different drugs estimated (Figure 5B). Antiviral efficacy scores of drugs were estimated by ranking similarities against the phase 2 group, indicating that amiloride, diltiazem, and the three microtubule inhibitors demonstrated relatively strong therapeutic effects (Figure 5C). The spatial distribution relationships of the drug treatment groups using t -distributed stochastic neighbor embedding (t -SNE) further illustrated the antiviral effects of the five molecules (Figure S12).

To validate the machine learning results, we initially assessed the effects of drugs on viral transport through fluorescence imaging. At 2 h post-infection, the five antiviral drug treatment groups identified by machine learning all exhibited a tendency to impede perinuclear virus enrichment, while other groups did not exhibit this trend (Figure S13). Immunofluorescence and $q\text{PCR}$ were subsequently employed to detect protein and RNA levels of progeny viruses following treatment with the five effective drugs (Figure 5D–G), with results consistent with machine learning output. Amiloride and diltiazem were originally approved as sodium channel inhibitors and calcium channel inhibitors for the treatment of conditions such as hypertension,^{40,41} but they have shown optimal antiviral effects. Vincristine and nocodazole exhibited the second-highest therapeutic efficacy by disrupting microtubule polymerization, essential for rapid viral movement. Conversely, the antiviral effect of paclitaxel was limited due to its inhibition of microtubule dissociation and maintenance of microtubule network integrity (Figure S14). These findings were consistent with those obtained using the SVIF-based machine learning approach, further validating its utility in antiviral drug screening and efficacy evaluation, underscoring the potential of SVIFs in assisting multidimensional database development.

Study Limitations. Our research findings indicated that the integration of SVT with machine learning presents a promising approach for the repurposing of antiviral drugs and the evaluation of their efficacy. However, our platform possesses

several limitations in light of the urgent need for rapid drug repurposing and efficacy assessment. First, while the platform demonstrated rapid and effective screening capabilities for drugs that significantly impact viral movement, it is less effective for drugs that possess antiviral properties but do not alter the viral dynamic behavior (e.g., drugs that impact viral replication and assembly). Second, while the method enables the swift screening of approved drugs, it can also be applied to candidate drugs or lead compounds. However, these compounds typically lack established concentration benchmarks, necessitating determination through additional experiments, which may extend the preparation time and limit the rapid applicability of this approach in certain cases. Moreover, the identification and assessment of antiviral drugs in this study are based on comparative analyses of viral movement characteristics, raising the possibility of establishing a comprehensive database for these. This database could be integrated into multidimensional big data analysis for drug screening. However, this study did not address potential issues with data reproducibility stemming from variability in experimental materials, nor did it incorporate additional data sources, such as transcriptomics, for multidimensional analysis.

CONCLUSIONS

While drug repurposing has proven to be a valuable strategy in the fight against emerging viruses, the current challenge lies in the lack of effective methods for intuitive visualization and immediate assessment of the effects of potential antiviral drugs. In this work, we developed a powerful platform that enables fast and efficient labeling of viruses, high-throughput SVT, construction of SVIFs, and machine-learning analysis for the prediction of antiviral drug efficacy in less than 90 min. We showed that QDs with a size smaller than 10 nm are suitable for capturing the intrinsic dynamic characteristics of a virus infection. Our study also demonstrated the potential of SVIFs to reflect perturbations experienced by viruses during dynamic infections. In addition, we highlighted the effectiveness of the logistic regression model in analyzing complex and similar experimental groups. Finally, we demonstrated the ability of SVIFs combined with machine learning for antiviral drug screening and efficacy assessment. This method provides visual evidence of the ability of drugs to combat viral infections in a remarkably short analysis time. Consequently, it holds significant promise for widespread application in addressing future outbreaks of emerging viruses.

METHODS

MDCK Cell Culture and Virus Preparation. MDCK cells (ATCC, CCL-34) were maintained in Dulbecco's modified Eagle's medium (DMEM; Gibco) supplemented with 5% fetal bovine serum (FBS; South American origin, PAN Biotech) in an environment with 5% CO₂ at 37 °C. The H9N2 strain of IAV (strain A/chicken/Hubei/01-MA01/1999) was cultured in 10 day-old specific pathogen-free chicken embryos at 37 °C and subsequently purified by sucrose gradient centrifugation as previously described.²⁰ Harvested viruses were stored at −80 °C for further use. For qPCR assay and immunofluorescence staining, viral infection occurred in virus culture medium composed of DMEM containing 0.3% bovine serum albumin (BSA) (SolarBio), 2 μg/mL tosylsulfonyl phenylalanyl chloromethyl ketone (TPCK)-trypsin (Sigma), and 1% penicillin–streptomycin.⁴² For virus tracking and titer detection after drug treatment, cells were pretreated with the drugs at the concentration indicated below at 37 °C for 30 min. DMSO 0.1%vol (Sigma), NH₄Cl 10 mM (Sigma), paclitaxel 50 μM (MCE), vincristine sulfate 20 μM (MCE), nocodazole 20 μM

(MCE), amiloride 16 μM (MCE), apigenin 20 μM (MCE), diltiazem 50 μM (MCE), Bafilomycin A1 20 μM (Cayman), and atorvastatin 20 μM (Glpbio).

Synthesis and Modification of CdZnSeS Quantum Dots. CdZnSeS QDs and OPA were synthesized according to the reported method.^{43,44} Oil-soluble QDs of different sizes were prepared by varying the ratio of Zn/S. Then, the hexane-dispersed QDs were washed with methanol and resuspended in chloroform. Subsequently, excess 3-MPA, DHLA, or OPA was dissolved in chloroform and slowly added to the QD solution. After sonication for several minutes, the solvent was removed by a rotary evaporator. The QDs were redissolved in Britton–Robinson buffer solution (pH 9.0) and subsequently purified through ultracentrifugation.

The OPA-coated QDs were further modified with PEG and SA. H₂N–PEG2000–COOH was covalently coupled to the surface of the OPA–QDs in the presence of 1-ethyl-3-(3-(dimethylamino)propyl) carbodiimide hydrochloride (EDC). OPA–QDs, EDC, and H₂N–PEG2000–COOH were dissolved in phosphate buffer solution (PBS) at a molar ratio of 1:3000:600 and reacted at room temperature for 2.5 h. Excess reactants were removed through repeated ultrafiltration (100 kDa, MWCO). The SA functionalization of the PEG–QDs followed a similar procedure, with the PEG–QDs, EDC, and SA reacting at a molar ratio of 1:2000:10. The obtained SA–QDs were dissolved in PBS and stored at 4 °C.

Transmission Electron Microscopy (TEM) Imaging. TEM images were acquired on a transmission electron microscope (JEOL, Japan, 80 kV). The oil-soluble QDs were dripped onto copper grids coated with Formvar film and then dried in a room temperature environment for TEM imaging without any staining.

Quantum Yield Measurement of QDs. The absolute photoluminescence quantum yield (PLQY) of the QDs was measured using a 500 nm xenon lamp as a light source, a visible PMT 900 (Edinburgh Instruments Ltd.) as a detector to collect the signal, and an integrating sphere (SM4 from Edinburgh Instruments Ltd.). First, the spectral signals of the blank samples were collected, where chloroform was used as the blank sample for the oil-soluble QDs and PBS for the water-phase QDs. The spectral signals of the QD solutions were measured under the same conditions. Finally, the PLQY of the QDs was obtained using the software provided with the Edinburgh Instrument.

Virus Labeling and Tracking. Purified viruses were incubated with EZ-link Sulfo-NHS-LC-Biotin (Thermo) at room temperature for 30 min. Following this, free biotin and virus aggregates were removed using NAP-5 gel filtration columns (GE Healthcare) and 0.22 μm-pore-size filters. Biotinylated viruses were then incubated with cells at 4 °C for 5 min, with subsequent washing to remove unattached viruses. Finally, 3 nM SA–QDs were added to label the attached viruses at 4 °C for 5 min, and the unbound QDs were washed away. For virus counting and labeling efficiency assays, purified viruses were incubated with 50 μM Syto 82 (Thermo) at room temperature for 2 h, followed by the removal of free Syto 82 using a NAP-5 column. For measuring the inhibition efficiency for viral transport, after internalization of QDs-labeled viruses, SA-Cy3 (1:5000 v/v) (Thermo) was applied to label non-internalized QDs-labeled viruses at 4 °C for 10 min. For SVT, the drug-treated cell samples were maintained in DMEM and imaged on a confocal microscope equipped with a cell culture system.

Viral Infectivity Assay. The viral infectivity of QDs-labeled H9N2 was examined by quantifying the hemagglutinin (HA) RNA level of the progeny H9N2 virus in MDCK cells by RT-qPCR. MDCK cells were seeded into 6-well plates at 5 × 10⁵ cells per well. After 24 h, the cells were incubated with QDs-labeled viruses and unlabeled viruses (MOI = 1), respectively, for 10 min at 4 °C for virus attachment. Then, the cells were washed and infected by viruses for 24 h at 37 °C. Following this, the cells were collected and lysed to extract RNA for RT-qPCR analysis.

Transfection of Cells. For tracking viral movement on microtubules, MDCK cells were transiently transfected with GFP-MAP4 using the JetPRIME transfection reagent (Polyplus) according to the relevant protocol. A final volume of 100 μL was obtained by diluting 2 μL of JetPRIME reagent and 1 μg of plasmid with JetPRIME buffer, followed by incubation for 10 min. The mixture was added to the cells

and incubated at 37 °C for 4 h before changing the medium. After 24 h of transfection, the dishes were used for imaging experiments.

Immunostaining. MDCK cells were pretreated with the indicated concentrations of drugs at 37 °C for 30 min and then incubated with viruses at 4 °C for 5 min. After that, the viruses were internalized into the cells in the virus culture medium containing the specified concentration of drugs at 37 °C under 5% CO₂ for 2 h. Subsequently, cells were fixed with 4% paraformaldehyde and permeabilized with 0.3% Triton X-100 for 15 min, respectively. Next, cells were blocked with 5% (w/v) BSA for 30 min and incubated with the rabbit monoclonal antibody against influenza virus nucleoprotein (1:300 v/v) (Sino Biological) for 1.5 h, followed by incubation with Dylight 649-labeled goat anti-rabbit IgG (1:300 v/v) (Abbkine) for 1 h. Finally, cells were incubated with 10 µg/mL Hoechst 33342 to label the cell nuclei and imaged on a confocal microscope. Fiji was used to quantify the fluorescence intensity of Dylight 649 within the cell nuclei.

Real-Time Reverse Transcription Quantitative PCR (RT-qPCR) Analysis. MDCK cells were seeded in 6-well plates at 5 × 10⁵ cells per well. After 24 h, cells were pretreated with the specified concentration of the drugs for 30 min, followed by infection of the viruses at MOI = 1 in the presence of the drug-supplemented virus growth medium. Cells were harvested at 2 and 24 h post-infection for RT-qPCR analysis. Then total cellular RNA was purified by using the MiniBEST Universal RNA Extraction Kit (Takara). One µg of total RNA was reverse-transcribed into cDNA using random hexamers according to the manufacturer's instructions (StarScript III All-in-one RT Mix with gDNA Remover, GenStar). Real-time quantitative PCR was performed using a 2× RealStar Power SYBR qPCR Mix (Low ROX) (GenStar) and an Applied Biosystems 7500 Fast real-time PCR system. Relative quantification was performed using the $\Delta\Delta C_T$ method with Gapdh serving as the endogenous control. The relative fold change in the cDNA levels of the H9N2 virus HA was calculated by normalization to the levels in the control cells. The primer sequence of HA (TSINGKE) was:

Forward: 5'-GTCAATGGTCAGCAGGGAAGAATTG-3'.

Reverse: 5'-GATTCTTCGGTGGCTCTCTCCTG-3'.

Fluorescence Imaging. The MDCK cells were cultured in 35 mm glass-bottom Petri dishes (NEST, China) for 24 h before fluorescence imaging. After incubation with the labeled viruses, the sample was imaged with a spinning-disk confocal microscope (Olympus IXplore SpinSR10) equipped with an online CO₂ culture system (INUB-WELSX-SET) and an sCMOS (Prime 95B) under a 100× objective. Hoechst 33342, GFP, Cy3/Syto 82/CdZnSeS QDs, and Dylight 649 were imaged by using 405, 488, 561, and 640 nm lasers and 447/60, 525/50, 617/73, and 685/40 nm emission filters, respectively. Fluorescence signals from different channels were imaged alternatively onto the sCMOS for simultaneous multicolor imaging.

Image and Statistical Analyses. The WCIF ImageJ software was utilized for analyzing the colocalization and intensity correlation of various fluorescent signals. The *t*M_r and *t*M_g values represented the colocalization percentage of red or green signals in the thresholded image, respectively. Statistical comparisons between fluorescent signals were determined using intensity correlation quotient values, with a range of +0.1 to +0.5 indicating strong covariance.⁴⁵ The product of differences from the mean image visualized the degree of colocalization of each pair of signals. Trajectories were constructed from the raw image sequence by using the TrackMate plugin in the Fiji ImageJ software. Fluorescence quantification was performed by using Fiji. In investigating the inhibitory effect of drugs on viral transport, ROIs were determined by reducing the cell outline to 1/3 and aligning it with the cell center. The one-way ANOVA was performed by using Graphpad Prism software to determine the significance of variations between groups. Significance level are stated as **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001.

Feature Description and Machine Learning. The SVIF comprises 17 features. The features T0, T1, T2, and T3 lifetime and meanSL indicate the variability in virus movement speed. The features such as kurtosis, Gaussianity, fractal dimension, and efficiency describe the directionality and asymmetry of viral transport trajectories. Meanwhile, trappedness, meanMSD, diffusion coefficient *D*, MSD

curve power-law fit quality *P*-value (*P*val), power-law scaling coefficient alpha, and MSDratio, are utilized to characterize the type of viral movement and trajectory filling properties. Lastly, feature *N* represents the duration of viral movement. A four-state HMM (T0, T1, T2, and T3) was used instead of a seven-state HMM to describe the viral transport process involving memoryless transitions between different states, which reduces the computational complexity while simplifying and ensuring a clear distinction between the fast and slow states of viral transport. The machine learning code was derived from the single-particle diffusion fingerprinting method⁴⁶ with an adjustment that utilizes the first quarter displacements of trajectories to compute MSD, while the formulas for calculating the other features remain unchanged.

Feature Visualization. The SVIFs for the 13 groups were preliminarily analyzed via parallel plots that plotted the 17 min to maximum normalized features for all trajectories. The mean, upper range, and lower range were also plotted to better visualize the feature characteristic. The mean value of each z-score normalized feature in each group was extracted and then mapped to a color scale in order to generate a heatmap of feature fingerprints that emphasizes the distinctions in the SVIFs among the groups. In addition, the SVIFs of drug treatment groups were projected into a two-dimension space using *t*-SNE for visual assessment of antiviral effects.

Similarity Assays. The Euclidean distance matrix was computed for each pair of groups, with each grid indicating the similarity between the specific two viral trajectories from the group pairs. The average of the Euclidean distance matrix was then calculated to represent the similarity between the two groups.³¹

Relative Antiviral Efficacy Assessment. Assessment of antiviral efficacy relied on the difference in the SVIFs between the treatment group and the control group. The average Euclidean distance of the SVIFs between the drug-treated and DMSO-treated phase 2 groups was calculated and normalized via min–max normalization, serving as the antiviral score. A higher score indicates a significant inhibitory effect of the corresponding drug on viral transport.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsnano.4c10136>.

Characterization of QDs, labeling efficiency evaluation, imaging condition optimization, representative images of viral movement, feature comparison among drug treatment groups, and analysis of drug inhibition on viral transport (PDF)

Typical trajectories of phase 2 (AVI)

Typical trajectories of phase 1 (AVI)

Typical trajectories of phase 0 (AVI)

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Notes

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